

PrepVI — Prototype Workflow Specification v2

1. Purpose and Prototype Scope

The prototype validates whether PrepVI can help a Student who does not know what to study move from a real Job Description (JD) to an explainable, personalized interview preparation path.

2. Prototype Narrative

Current-state story

A Student is preparing for an internship or job application but does not know which topics from the Job Description deserve the most attention. The Student searches for interview questions from different sources, takes fragmented notes, and has difficulty deciding what to practice first.

Future-state story

The Student uploads or pastes a real Job Description. PrepVI extracts the text, lets the Student review and correct it, maps the relevant requirements to the existing interview taxonomy, and produces an explainable question set. The Student can then open a recommended question and begin practicing immediately.

If additional help is needed, the Student can continue into the secondary mentoring flow to find a Mentor, schedule a session, receive structured feedback, and return to the Question Bank with recommended next actions.

3. Core Prototype

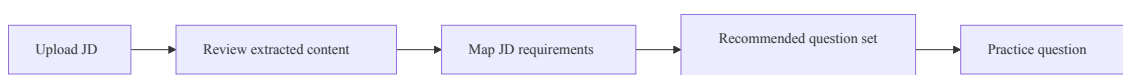
3.1 Core Hypothesis

A Student who does not know what to study can use a Job Description to obtain an understandable and useful interview preparation path.

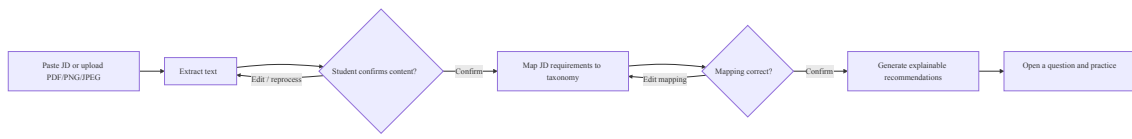
The Core Prototype must validate three things:

1. The Student understands what information PrepVI extracted from the JD.
2. The Student understands and can correct how JD requirements are mapped to interview topics.
3. The Student understands why the resulting questions are recommended and can start practicing from them.

3.2 Core Flow



Alternative correction loop:



3.3 Upload Job Description

Prototype frame:

Purpose: Reduce the starting barrier for a Student who does not know what to study.

Prototype behavior

- Allow the Student to paste JD text or upload one supported document/image.
- Supported input may include PDF, PNG, and JPEG.
- Show a preview before submission where applicable.
- Explain supported size/page constraints.
- Show clear validation when the file is unsupported or unreadable.
- Include a privacy notice encouraging the Student to remove unnecessary personal information.
- Provide a clear CTA such as **Extract content**.
- Prevent accidental duplicate submission in the interaction design.

Validation question

Can the Student understand how to provide a JD without assistance?

3.4 Review Extracted Content

Prototype frame:

So sánh với ảnh gốc và chỉnh sửa nếu cần. Các đề xuất tiếp theo sẽ dựa trên phiên bản này.

⚠️ Có 1 đoạn cần kiểm tra
Những dòng được đánh dấu vàng có thể chưa được đọc chính xác – nhấn "Sửa" để áp dụng gợi ý.

Ảnh gốc + - □ Nội dung trích xuất Nhấp vào dòng để chỉnh sửa

Frontend Intern

Requirements

- Solid knowledge of JavaScript and ES6+
- Basic understanding of React and component lifecycle
- Familiarity with REST API and HTTP methods
- Git knowledge and version control workflows
- Good communication skills

Nice to have

- Experience with TypeScript
- Basic understanding of CSS-in-JS solutions

← Trang 1/1 →

ⓘ Các đề xuất câu hỏi tiếp theo sẽ dựa trên phiên bản bạn xác nhận ở đây. Bạn có thể chỉnh sửa bằng cách nhấp vào từng dòng.

< Quét lại Xác nhận nội dung → ?

Purpose: Give the Student control over the input before recommendations are generated.

Prototype behavior

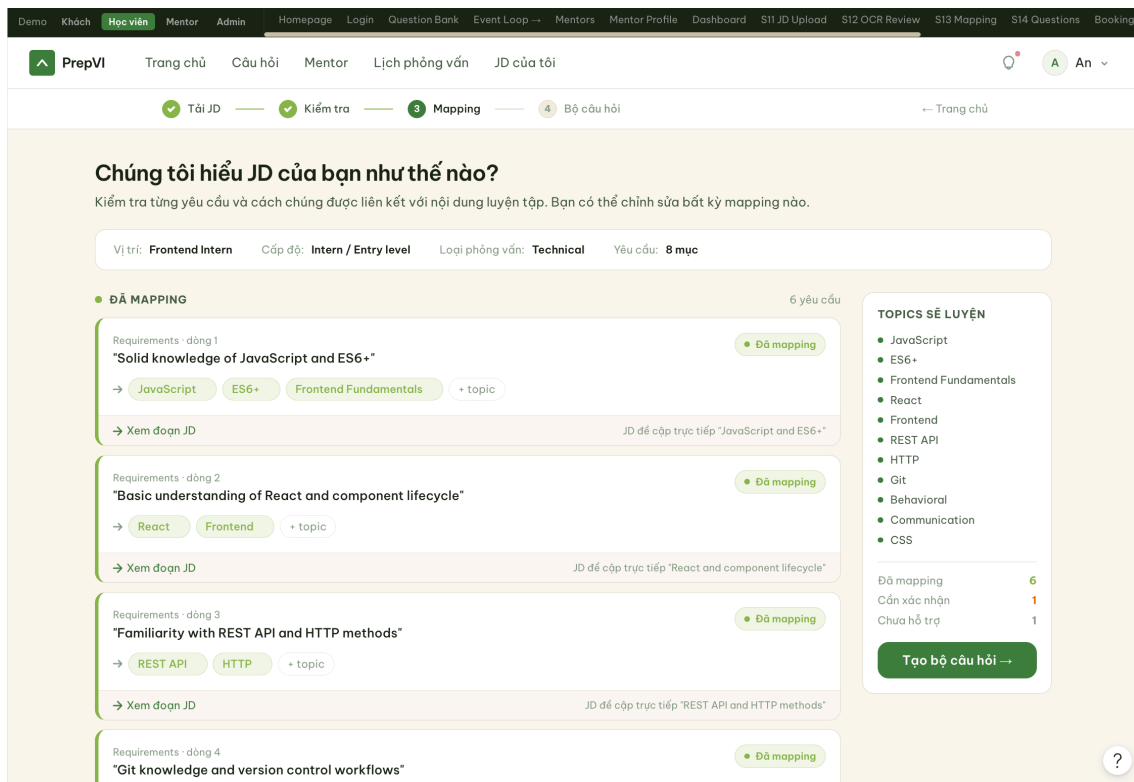
- Show the source JD next to the extracted text where practical.
- Make extracted text editable.
- Mark uncertain or unreadable segments rather than silently inventing content.
- Allow the Student to correct the content.
- Provide **Reprocess** and **Confirm content** actions.
- Do not allow mapping to continue when the extracted content is empty or clearly insufficient.

Validation question

Can the Student detect and correct important extraction mistakes before the system uses the JD?

3.5 JD Requirement Mapping

Prototype frame:



Purpose: Explain how PrepVI interprets the JD before recommending questions.

Prototype behavior

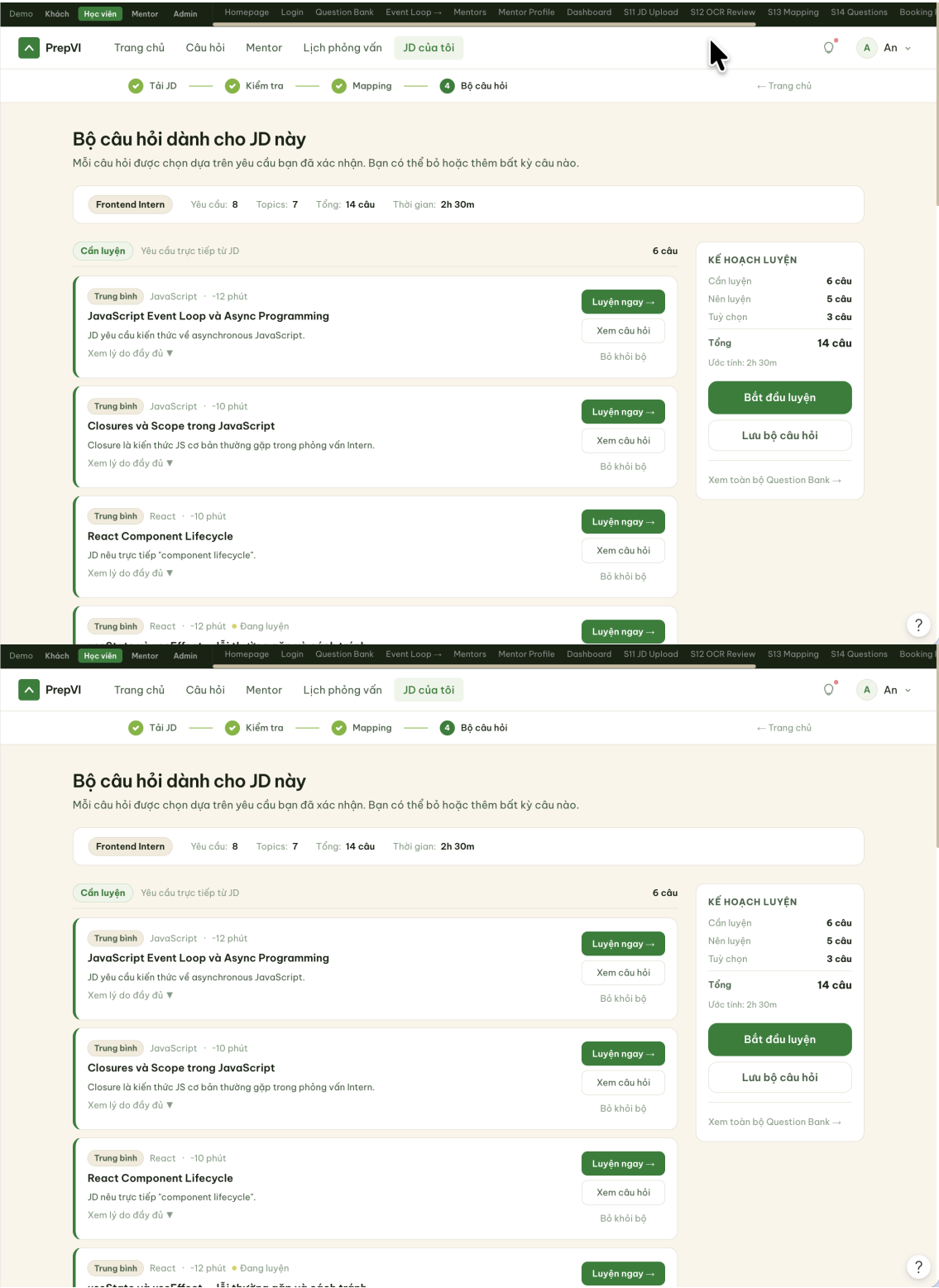
- Extract meaningful requirements such as position, seniority, skills, topics, and interview context.
- Map each relevant JD requirement to the existing question taxonomy.
- Show the source JD fragment that led to each mapping.
- Distinguish between:
 - mapped,
 - needs confirmation,
 - unsupported / no taxonomy match.
- Allow the Student to edit, remove, or reject incorrect mappings.
- Avoid adding skills that are not supported by the JD.
- Enable **Create question set** only when at least one useful mapping exists.

Validation question

Does the Student understand how PrepVI interpreted the JD, and can the Student correct a wrong interpretation?

3.6 Recommended Question Set

Prototype frames:



Purpose: Turn JD requirements into a concrete and explainable preparation starting point.

Prototype behavior

- Group recommended questions by priority, for example:
 - Must practice,
 - Should practice,
 - Optional.
- Show topic, difficulty, and estimated effort where useful.
- Every recommendation should include a clear reason tied to a mapped JD requirement.
- Allow the Student to remove irrelevant questions.
- Allow the Student to add questions from the Supporting Question Bank.
- Provide **Practice now** or **Start practicing**.
- Preserve the connection between the question and the JD reason when entering practice.
- Show an empty state if no useful recommendations are available.

Validation question

Can the Student explain why a question was recommended and decide what to practice next?

3.7 Practice Question

The screenshot shows the PrepVI web application interface. At the top, there's a navigation bar with links like 'Trang chủ', 'Câu hỏi', 'Mentor', 'Lịch phỏng vấn', and 'JD của tôi'. The main content area is titled 'Question Bank / JavaScript' and features a question titled 'Giải thích Event Loop trong JavaScript và cách microtask khác macrotask.'. Below the title, there are buttons for 'Chưa luyện', 'Đang luyện', 'Tự tin', and 'Đang luyện'. The question content consists of four numbered prompts: 1. Giải thích call stack, task queue và microtask queue; 2. Phân biệt Promise callbacks (microtask) với setTimeout (macrotask); 3. Mô tả thứ tự thực thi với ví dụ cụ thể; 4. Đề cập đến requestAnimationFrame nếu biết. To the right, there's a sidebar with a section 'Muốn luyện câu này với mentor?' and a button 'Tìm mentor cho chủ đề này'. The bottom of the page shows related questions like 'React reconciliation hoạt động như thế nào? Virtual DOM là gì?' and 'CSS specificity hoạt động như thế nào? Giải thích với ví dụ.'.

Purpose: Complete the Core Prototype by proving that the recommendation leads to an actionable practice step.

Prototype behavior

- Show question content and relevant context.

- Show answer criteria, hints, or guidance where appropriate.
- Preserve provenance or recommendation reason when useful.
- Allow a simple practice state such as:
 - Not started,
 - Practicing,
 - Confident.
- Avoid presenting one rigid "correct answer" for behavioral questions.
- Allow the Student to continue to the mentor flow when human feedback is needed.

Validation question

Can the Student move from recommendation to actual practice without losing context?

4. Supporting Prototype

S02 — Question Bank

Prototype frame:

The screenshot shows a web application interface for a 'Question Bank'. The top navigation bar includes links like Demo, Khách, Học viên, Mentor, Admin, and various tool links. The main header shows the user's profile (PrepVI) and navigation tabs (Trang chủ, Câu hỏi, Mentor, Lịch phỏng vấn, JD của tôi). The 'Câu hỏi' tab is active.

The 'Question Bank' section has a search bar and filters for 'Vị Trí', 'Chủ Đề', 'Loại PV', 'Độ Khó', and 'Trạng Thái'. The current filters are 'Frontend Intern' and 'JavaScript'. There are 8 questions listed, each with a title, tags, and a 'Sắp xếp' dropdown set to 'Phù hợp nhất'.

The first question is 'Giải thích Event Loop trong JavaScript và cách microtask khác macrotask.' with tags 'JavaScript', 'Frontend', 'Trung bình', and 'Đang luyện'. It has 312 practice attempts. The second question is 'React reconciliation hoạt động như thế nào? Virtual DOM là gì?' with tags 'React', 'Frontend', 'Trung bình', and 'Tự tin'. It has 245 practice attempts. The third question is 'CSS specificity hoạt động như thế nào? Giải thích với ví dụ.' with tags 'CSS', 'Frontend', 'Dễ', and 'Tự tin'. It has 198 practice attempts. The fourth question is 'Thiết kế một hệ thống rút gọn URL (URL shortener) như bit.ly.' with tags 'System Design' and 'Khó'. It has 421 practice attempts. The fifth question is 'Kể về một lần bạn xử lý xung đột với đồng nghiệp. Bạn đã làm gì?' with tags 'Behavioral' and 'Trung bình'. It has 156 practice attempts.

On the right, a sidebar titled 'MỤC TIÊU HIỆN TẠI' shows progress for 'Frontend Intern' (Phỏng vấn trong 18 ngày). It lists progress for 'JavaScript' (7/20), 'React' (3/15), and 'Behavioral' (1/12). Below this is a button 'Luyện câu hỏi Behavioral →'.

The Question Bank supports the Core Prototype but is not the main hypothesis being validated.

Purpose

- Let the Student inspect, expand, or replace JD-based recommendations.

- Provide a fallback when the taxonomy mapping does not produce enough useful results.

Prototype behavior

- Search by keyword.
- Filter by Position, Topic, Interview Type, and Difficulty.
- When opened from S14, preserve the filters derived from JD mappings.
- Let the Student remove individual JD-derived filters.
- Provide a meaningful zero-result state.
- Avoid duplicate rendering of a question that has multiple tags.
- Support clear pagination or load-more behavior.

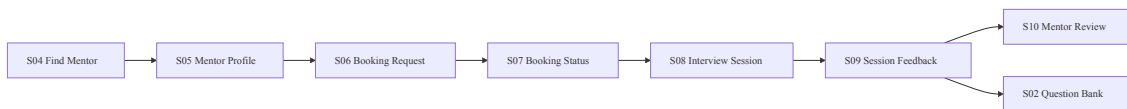
Supporting validation question

Can the Student recover when recommendations are incomplete and still find relevant questions manually?

5. Secondary Flow

The Secondary Flow validates the mentoring and feedback experience. It is useful to the overall product, but it is separate from the main **JD** → **personalized preparation** hypothesis.

5.1 Student Secondary Flow



S04 — Find Mentor

Prototype frame:

- Show public bio, expertise, verification status, service scope, ratings, and availability.
- Show timezone clearly.
- Explain that the actual meeting may use an external provider.

S06 — Booking Request

Gửi yêu cầu đặt lịch

THÔNG TIN BUỔI LUYỆN TẬP

Vị trí đang ứng tuyển *

vd. Frontend Intern tại Shopee

Loại phỏng vấn *

Chọn loại phỏng vấn

Mục tiêu buổi luyện tập *

vd. Tôi muốn luyện câu hỏi về Event Loop và cải thiện cách cấu trúc câu trả lời.

Nội dung muốn luyện

vd. Event Loop, React Hooks, CSS Flexbox

Ghi chú thêm (tùy chọn)

Bất kỳ thông tin nào giúp mentor chuẩn bị tốt hơn.

TÓM TẮT ĐẶT LỊCH

Nguyễn Minh Tuấn
Senior Frontend Engineer

Ngày: Thứ Ba, 18 tháng 3, 2025
Giờ: 09:00 sáng
Múi giờ: GMT+7 - Hồ Chí Minh
Thời lượng: 60 phút
Hình thức: Video call

Chính sách

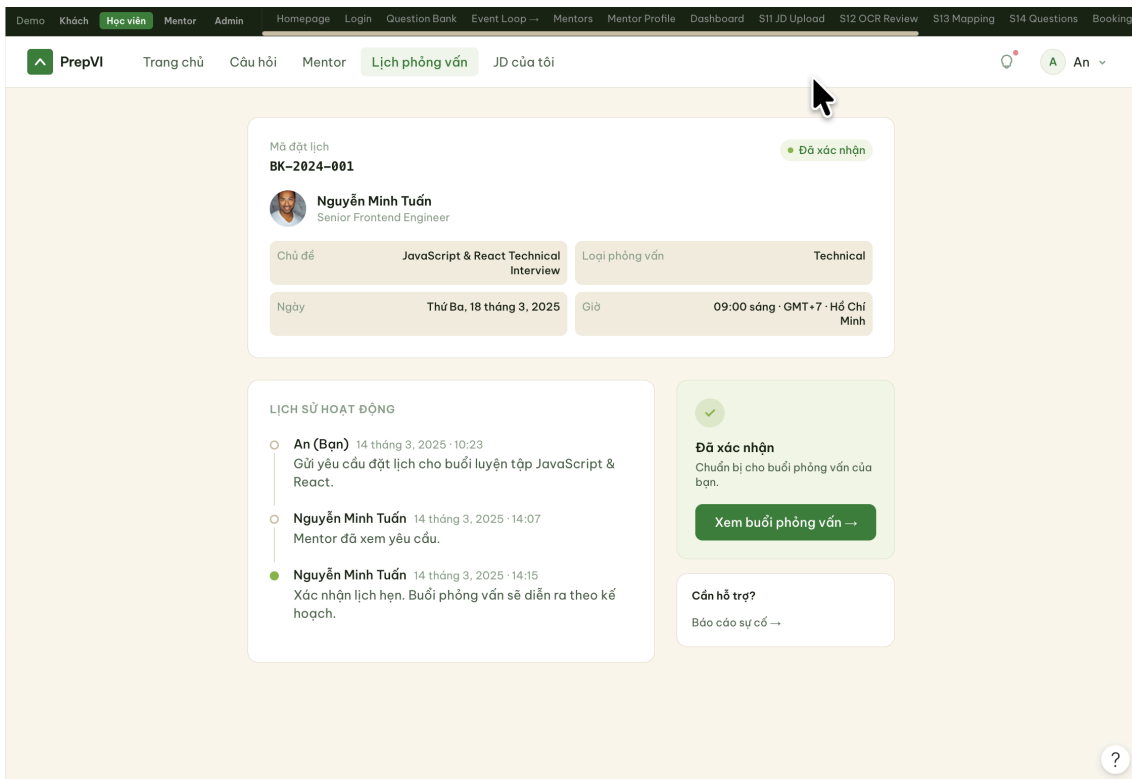
Chính sách hủy lịch
Bạn có thể hủy lịch ít nhất 24 giờ trước buổi phỏng vấn. Hủy muộn hơn sẽ không được hoàn lịch.

Chính sách no-show
Nếu bạn không xuất hiện tại buổi phỏng vấn mà không có lý do chính đáng, bạn sẽ bị đánh dấu là no-show và không thể đặt lịch lại trong 7 ngày.

- Show fixed Mentor and slot summary.
- Require interview goal / target role / practice focus.
- Allow recommended questions or topics to be attached.
- Show relevant cancellation/no-show policy before submission.

S07 — Booking Status

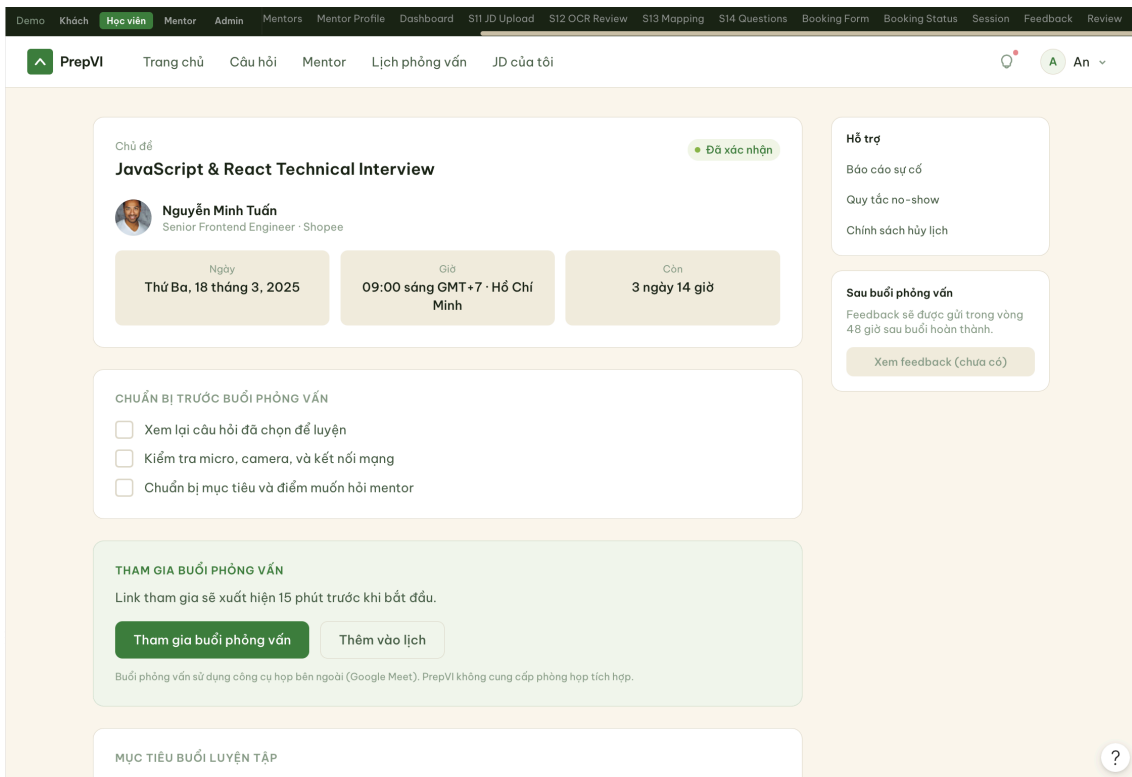
Prototype frame:



- Show Pending, Confirmed, Reschedule proposed, Rejected, and Cancelled states.
- Make the next valid action clear.

S08 — Interview Session

Prototype frame:



- Show goal, topic, Mentor, local time, and external meeting information when appropriate.
- Keep actions consistent with booking state.

S09 — Session Feedback

Prototype frame:

Demo Khách **Học viên** Mentor Admin Mentors Mentor Profile Dashboard S11 JD Upload S12 OCR Review S13 Mapping S14 Questions Booking Form Booking Status Session Feedback Review

PrepVI Trang chủ Câu hỏi Mentor Lịch phỏng vấn JD của tôi

Feedback buổi luyện tập

Mentor: **Nguyễn Minh Tuấn** · Technical Interview · 18 tháng 3, 2025

NHẬN XÉT TỔNG QUAN

"Bạn đang làm tốt phần kiến thức, nhưng cần cấu trúc câu trả lời rõ hơn trước khi đi vào chi tiết."

ĐÁNH GIÁ THEO TIÊU CHÍ
Điểm phản ánh chất lượng buổi luyện, không phải năng lực tuyệt đối của bạn.

Kiến thức kỹ thuật **4/5**
Hiểu sâu về Event Loop và Promises. Có thể giải thích microtask vs macrotask một cách chính xác.
Tốt
[Trả lời đúng về thử tự thực thi của Promise.then và setTimeout.](#)

Cấu trúc câu trả lời **3/5**
Câu trả lời đôi khi bắt đầu quá chi tiết trước khi xác định context. Cần lập outline trước.
Đạt yêu cầu
[Câu hỏi về reconciliation: bắt đầu giải thích Fiber ngay mà chưa giới thiệu Virtual DOM.](#)

Giao tiếp **4/5**
Giải thích rõ ràng, dùng ví dụ thực tế tốt. Cần chủ động check-in với người phỏng vấn.
Tốt
[Ví dụ closure với counter function rất trực quan.](#)

Xử lý câu hỏi tiếp theo **3/5**

- Show structured rubric feedback.
- Include strengths, weaknesses, evidence, and next actions.
- Link next actions back to relevant topics or questions.
- Keep feedback private to appropriate actors.

S10 — Mentor Review

Prototype frame:

Demo
Khách
Học viên
Mentor
Admin
Mentors
Mentor Profile
Dashboard
S11 JD Upload
S12 OCR Review
S13 Mapping
S14 Questions
Booking Form
Booking Status
Session
Feedback
Review

PrepVI
Trang chủ
Câu hỏi
Mentor
Lịch phỏng vấn
JD của tôi
An

Buổi luyện tập vừa rồi thế nào?

Đánh giá chỉ dành cho buổi đã hoàn thành.



Nguyễn Minh Tuấn
 JavaScript & React Technical Interview · Thứ Ba, 18 tháng 3, 2025

Đánh giá tổng thể

★ ★ ★ ★ ★

Nhận xét của bạn

Buổi luyện tập có ích như thế nào? Mentor giải thích rõ không? Bạn học được gì?

Hướng dẫn viết đánh giá

- Tập trung vào chất lượng buổi học, không phải ý kiến cá nhân.
- Phản hồi cụ thể sẽ giúp mentor cải thiện và giúp học viên khác chọn mentor phù hợp.
- Đánh giá có thể bị ẩn nếu vi phạm chính sách cộng đồng.

Đánh giá của bạn sẽ được kiểm duyệt và hiển thị công khai trên hồ sơ mentor sau khi được duyệt.

Gửi đánh giá

- Allow rating and comment after an eligible completed booking.
- Explain review visibility/moderation where relevant.

7. States to Prototype

Prototype states should be prioritized by how strongly they affect user understanding of the validated flow.

7.1 Core states — Must Prototype

State	Relevant screens	Why
Loading / processing	S11–S14	User must understand that work is in progress
Empty input	S11–S12	Prevents an invalid start
Validation error	S11–S13	Requirement clarification
OCR / extraction uncertainty	S12	Lets the Student correct input
Unsupported content	S11–S12	Provides recovery path
No taxonomy match	S13	Prevents false interpretation
Incorrect mapping	S13	Directly validates user control
No recommended question	S14	Provides recovery without losing prior work
Zero-result Question Bank	S02	Supports manual fallback

7.2 Secondary states — Prototype when relevant

- No Mentor matching filters.
- No available slot.
- Booking Pending / Confirmed / Rejected / Cancelled.
- Reschedule proposed.
- Feedback not yet available.

7.3 Operational states — Document but do not overclaim

- Permission denied.
- Concurrent slot conflict.
- Provider failure.
- Offline/timeout retry.
- Audit/history behavior.

These may appear in the prototype as interaction states, but they must not be treated as proof that production security, concurrency, reliability, or audit behavior has been implemented correctly.

8. Prototype Test Plan

8.1 Core Prototype Test Tasks

Task	Persona	Success criterion
Upload or paste a JD	Student	Completes without assistance
Review extracted text	Student	Detects and corrects an important extraction mistake
Review JD mappings	Student	Understands mapped / uncertain / unsupported items
Correct a wrong mapping	Student	Changes or removes it successfully
Review recommended questions	Student	Can explain why at least one question was recommended
Start practicing	Student	Opens a recommended question without losing JD context
Recover from no useful recommendation	Student	Reaches S13 or S02 and finds another path

8.2 Supporting / Secondary Tasks

Task	Persona	Success criterion
Find a Front-end / JavaScript question	Student	Finds a relevant result with no assistance
Find a Mentor	Student	Selects suitable expertise and timezone
Submit a booking	Student	Completes and understands Pending state
Respond to a booking	Mentor	Chooses a valid action and understands its effect
Submit structured feedback	Mentor	Provides strengths, improvement areas, and next actions
Review a Mentor application	Admin	Makes a decision with reason

8.3 Evidence to Collect

Collect:

- task completion,
- time on task,
- errors,
- confusion points,
- confidence,
- qualitative comments,
- mapping corrections,
- recommendation usefulness.

Recommended prototype acceptance targets may include:

- Student core task completion $\geq 80\%$,
- most critical extraction mistakes can be corrected before mapping,
- Students can explain the reason behind a recommended question,
- average perceived usefulness of the generated preparation set $\geq 4/5$.

These targets are prototype acceptance goals, not population-level statistical proof.

9. Prototype vs Proof of Concept

The terms **Prototype** and **Proof of Concept (PoC)** should be kept distinct.

Prototype

Validates how users understand and interact with the proposed solution.

Example:

Upload JD → Review → Mapping → Recommendations → Practice

Typical evidence:

- usability observation,
- user feedback,
- task completion,
- confusion points,
- requirement clarification.

Technical PoC

Validates whether a technical idea is feasible enough to continue exploring.

Example:

Document → Text extraction/OCR → Requirement parsing → Taxonomy mapping → Question
retrieval

A technical PoC may exist without a polished UI.

The UI workflow in this document should primarily be treated as the **Prototype**, while OCR, taxonomy mapping, and recommendation experiments may be evaluated separately as technical PoCs.